Zhigen Zhao

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Fox School of Business
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EDUCATION

- Ph.D. in Mathematics, 2009, **Cornell University**, Ithaca, NY
  Ph.D. Dissertation: *Decision Approach and Shrinkage Confidence Intervals*
- B.S. in Mathematics, 2003, **Nankai University**, Tianjin, China

EMPLOYMENT

- Professor, Department of Statistics, Operations and Data Science, Fox School of Business and Administration, Temple University, 2026-present.
- Department Chair, Department of Statistics, Operations and Data Science, Fox School of Business and Administration, Temple University, 2026-present.
- Charles E. Beury Senior Research Fellow, Fox School of Business and Administration, Temple University, 2026-present.
- Associate Professor, Department of Statistics, Operations and Data Science, Fox School of Business and Administration, Temple University, 2021-2026.
- Associate Professor, Department of Statistical Science, Fox School of Business and Administration, Temple University, 2016-2021.
- Associate Professor (secondary appointment), Center for Data Analytics and Biomedical Informatics, Temple University, 2016-2019.
- Charles E. Beury Research Fellow, Fox School of Business and Administration, Temple University, 2016-2026.
- Visiting Scholar, Department of Statistics, Harvard University, 2019.
- Visiting Scholar, Department of Mathematics, Cornell University, 2010.
- Assistant Professor, Department of Statistics, Fox School of Business and Administration, Temple University, 2009-2016.

HONORS & AWARDS

- Co-Editor-in-Chief of *Communications in Statistics, Theory and Methods.* 2025-Present
- Faculty Fellowship Program in Israel. 2024
- Elected Member, International Statistical Institute (ISI). 2023

- Dean's Research Honor Roll, Fox School of Business Management, Temple University. 2014
- IMS Laha Travel Award. 2009

GRANT

1. Principal Investigator. STEM Research Funds. *Multivariate Empirical Bayes Shrinkage: Theory, Optimality, and Modern Applications.* 2026-2027. $75,000.
2. Principal Investigator. National Science Foundation. *Collaborative Research: Multiple Hypothesis Testing on the Regression Analysis.* 2023-2027. NSF-DMS-2311216. $197,009.
3. Principal Investigator. National Science Foundation. *BIGDATA: Collaborative Research: F: Statistical Theory and Methods beyond the Dimensionality Barrier.* 2016-2020. NSF-IIS-1633283. $250,000.
4. Co-Principal Investigator. Department of Transportation of State Pennsylvania. *Winter Roadway Maintenance Material Enhancers (Field) Evaluation.* 2016-2018. $298,000.
5. Principal Investigator. National Science Foundation. *Bayesian Decision Theoretic Methods for Some High-Dimensional Multiple Inference Problems.* 2012-2015. NSF-DMS-1208735. $174,976.

RESEARCH CONTRIBUTIONS

Manuscripts Under Review

1. Zhao, Z. and Sugasawa, S. (2026) Nonparametric f-Modeling for Empirical Bayes Inference with Unequal and Unknown Variances.
2. Zhang, K., Zhao, Z., and Zhou, W. (2024) BEAUTY Powered BEAST. Under review at *Statistical Science.*
3. Zhao, Z. and Xing, X. (2024) On the testing of multiple hypotheses in sliced inverse regression. Under review at *Electronic Journal of Statistics.*
4. Liang, Z. , Liu, J., Wang, X. and Zhao, Z. (2024) Bayesian Analysis of Multiple Index Additive Models. Under review at *Statistical Analysis and Data Mining.*
5. Zhang, W., Zhao, Z., Li, Y., Baiocchi, M., and Zhang K. (2025) SorBET: A Fast and Powerful Algorithm to Test Dependence of Variables. Under review at *Journal of Computational and Graphical Statistics.*
6. Zhang, W., Ni, H., Liu, Y., Zhang, K., and Zhao, Z. (2025) Regularizing BELIEF with Sequence Lasso. Under review at *Journal of the American Statistical Association.*

Manuscript in Preparation

1. Liu, L., Wang, X., Li, X. and Zhao, Z. (2025) The Impact of Leisure Travel on Loneliness among Older Adults in China and the United States.
2. Zu, T., Ji, P. and Zhao, Z. (2024) Two-stage Filter: an Optimal Multiple Testing Method in High-Dimensional Regression.

3. Yu, Y., Zu, T. and Zhao, Z. (2024) FDR control for high dimensional quantile variable selection.

4. Zhao, Z., Y. Tian, K. Zhang, and W. Zhou (2024). Test of exchangeability under binary expansion.

5. Ellingworth, A., Ghosh, D. , Zhao, Z. and Zhou, W. (2024) Reproducible or not: a data adaptive nonparametric procedure to define and assess reproducibility across high-throughput studies.

Publications

1. Yan, L., Zhao, Z., Zhang, H. (2026) Model-free multiple testing for matrix-valued predictors with false discovery control. *Journal of Computational and Graphical Statistics.* Available on line.

2. Sanni, S., Yu, Y. and Zhao, Z. (2026) Ultra-high dimensional threshold selection for quantile mirror feature screening with FDR error rate control: A case study on high blood pressure analysis. Biometrics. Volume 82, Issue 1.

3. Zhao, Z. (2025) Discussion on "Data Fission: Splitting a Single Data Point". *Journal of the American Statistical Association.* Volume 120, Issue 549, 178-179.

4. Liao, Y., Xiang, Y., Zhao, Z. and Ai, D. (2025) Bayesian Mixed Effect Higher-Order Hidden Markov Models with Applications to Predictive Healthcare Using Electronic Health Records. *IISE Transactions.* Volume 57, Issue 2, 186-198.

5. Zhao, Z., Wang, T., and Ji, B. (2024) Randomized Multiarm Bandits: An Improved Adaptive Data Collection Method. *Statistical Analysis and Data Mining.* Volume 17, Issue 2. E11681.

6. Xing, X., Zhao, Z. and Liu, J. (2023) Controlling False Discovery Rate using Gaussian Mirrors. *Journal of the American Statistical Association.* Volume 118, Issue 541, 222-241.

7. Kwon, Y., Zhao, Z. (2023) On F-modelling based empirical Bayes Estimation of Variances. *Biometrika.* Volume 110, Issue 1, March 2023, 69–81.

8. Sanat K. Sarkar, Zhao, Z. (2022) Local False Discovery Rate Based Methods for Multiple Testing of One-Way Classified Hypotheses. *Electronic Journal of Statistics.* Vol. 16, Issue 2, 6043-6085.

9. Zhao, Z. (2021) Where to find needles in a haystack? *TEST.* Vol. 31, Issue 1, 148-174.

10. Lin, Q., Zhao, Z., and Liu, J. (2021) Global testing under the sparse alternatives for single index models. *Festschrift in Honor of R. Dennis Cook.*

11. Lin, Q., Zhao, Z., and Liu, J. (2019) Sparse Sliced Inverse Regression via Lasso, *Journal of the American Statistical Association.* Vol. 114, Issue 528, 1726-1739.

12. Zhang, J., Zhao, Z. , Zhang, K., and Wei, Z. (2019) A Feature Sampling Strategy for Analysis of High Dimensional Genomic Data. *IEEE/ACM Transactions on Computational Biology and Bioinformatics.* Vol. 16, No. 2, 434-441.

13. Lin, Q., Zhao, Z., and Liu, J. (2018) On consistency and sparsity for sliced inverse regression in high dimension. *Annals of Statistics.* Vol. 46, No. 2, 580-610.

14. Anis, M., Zhao, Z. , Khurana, J., Krynetskiy, E. and Soliman, A. (2018) Determining candidate single nucleotide polymorphisms in acquired laryngotracheal stenosis. *The Laryngoscope.* Vol. 128, Issue 3, 111-116.

15. Ni, H., Qin, J., Zhou, L., Zhao, Z., Wang, J., Hou, F. (2017). Network analysis in detection of early-stage mild cognitive impairment. *Physica A: Statistical Mechanics and its Applications.* Vol. 478, 113-119.

16. Chang, Y. W., Tsong, Y., and Zhao, Z. (2017) Sample size determination for a three-arm equivalence trial of Poisson and Negative Binomial responses. *Journal of Biopharmaceutical Statistics.* Vol. 27, Issue 2, 239-256.

17. Liu, Y., Sarkar, S. K., and Zhao, Z. (2015) A new approach to multiple testing of grouped hypotheses. *Journal of Statistical Planning and Inference.* Vol. 179, 1-14. (One of the most downloaded articles from Journal of Statistical Planning and Inference in the last 90 days.)

18. He, L., Sarkar, S. K., and Zhao, Z. (2015) Capturing the severity of Type II errors in high-dimensional multiple testing. *Journal of Multivariate Analysis.* Vol. 142, 106-116.

19. Zhao, Z. and Sarkar, S. K. (2015) A Bayesian approach to construct multiple confidence intervals of selected parameters with sparse signals. *Statistica Sinica.* Volume 25, Number 2, 725-742.

20. Clements, N., Sarkar, S. K., Zhao, Z. and Kim, D. (2014) Applying multiple testing procedure to detect changes in East African vegetation. *Annals of Applied Statistics.* Volume 8, No.1, 286-308.

21. Anis, M., Zhao, Z., Khurana, J., Krynetskiy, E. and Soliman, A. (2014) Translational genomics of acquired laryngotracheal stenosis. *The Laryngoscope.* Vol. 124, Issue 5, 175-179.

22. Chang, Y. W., Tsong, Y., Dong, X. and Zhao, Z. (2014) Sample size determination for a three-arm equivalence trial of normally distributed responses. *Journal of Biopharmaceutical Statistics.* Vol. 24, Issue 6, 1190-1202.

23. Zhao, Z., Wang, W. and Wei, Z. (2013) An empirical Bayes testing procedure for detecting variants in analysis of next generation sequencing data. *Annals of Applied Statistics.* Volume 7, No.4, 2229-2248.

24. Hwang, J. T., Zhao, Z. (2013) Empirical Bayes confidence intervals for selected parameters in high dimensional data. *Journal of the American Statistical Association.* Volume 108, Issue 502, 607-618.

25. Zhao, Z. and Hwang, J. T. (2012) Empirical Bayes FCR controlling confidence intervals. *Journal of the Royal Statistical Society, Series B*, Volume 74, Issue 5, 871-891.

26. Zhao, Z. (2010) Double shrinkage empirical Bayesian estimation for unknown and unequal variances, *Statistics and Its Interface.* Volume 3, 533-541.

27. Hwang, J. T., Qiu, J. and Zhao, Z. (2009) Empirical Bayes confidence intervals shrinking both means and variances. *Journal of the Royal Statistical Society, Series B.* Volume **71**, Issue 1, 265-285.

PATENT

1. Methods, systems, and computer readable media for non-parametric dependence detection using bitwise operations in a computing system. 2019.

TEACHING

Graduate Course

- Stat8112, Statistical Methods in Business Research I (Average SFF Q8/Q11 and eSFF Q3/Q4 scores: 4.2, 4.2).
- Stat8113, Statistical Methods in Business Research II (Average SFF Q8/Q11 and eSFF Q3/Q4 scores: 4.5, 4.5).
- Stat8003, Statistical Methods I (Average SFF Q8/Q11 scores: 4.7, 4.5).
- BA9105, Econometrics I (Average SFF Q8/Q11 scores: 4.5, 4.3).
- Stat8106, Generalized Linear Models I (Average SFF Q8/Q11 scores: 4.8, 4.9).
- Stat8002, Probability Theory II (Average SFF Q8/Q11 scores: 4.4, 4.3).
- Stat8001, Probability Theory I (Average SFF Q8/Q11 scores: 4.6, 4.8).

Undergraduate Course

- Stat3502, Regression and Predictive Analytics (Average eSFF Q3/Q4 scores: 3.8, 3.8).
- Stat3503/Stat8109, Intermediate Business Statistics (Average SFF Q8/Q11 scores: 4.0, 4.2).
- Stat2103, Business Statistics (Average SFF Q8/Q11 scores: 3.8, 3.7).
- Stat2512/3507, Intermediate Statistics (Average SFF Q8/Q11 and eSFF Q3/Q4 scores: 4.0, 4.0).
- Stat2102, Selected Statistics Applications in Business (Average SFF Q8/Q11 scores: 4.3, 4.1).

MENTORSHIP

Ph.D. Dissertation Chair

- Isaac Lim. 2025 - present.
- Kyle Chen. 2023 - present.
- Jian Sun. 2019 - present.
- Yu Tian. 2017 - 2025. *Test of Exchangeability under Binary Expansion.*
- Tong Wang. 2017 - 2022. *Resampling-based Bias Adjustment for Adaptively Collected Data.*
- Zhengkang Liang. 2018 - 2022. *On the Bayesian Multiple Index Models.*

- Yeil Kwon. 2013-2018. *Nonparametric empirical Bayes simultaneous estimation for multiple variances.*
- Yanping Liu. 2009-2016. *New Approaches to Multiple Testing of Grouped Hypotheses.* Co-chaired with Dr. Sanat K. Sarkar.
- Victoria (Yu-Wei) Chang. 2009-2014. *Sample Size Determination for a Three-arm Biosimilar Trial.*

Ph.D. Dissertation Committee

• Erya Ouyang • Peiyu Liang • Yufan Dong • Dan Luo • Hailey Park • Mengtian Li • Lanyu Lei • Michael D. Power • Shinjini Nandini • Chao Han (Computer Science, Temple University) • Yanhui Xu • Zeda Li • Scott Bruce • Jie Zhang (Computer Science, NJIT) • Jelena Gligorijevic • Hang Kim • Kun Tang • Jing Xiao• Bu Hyoung Lee• Yiyong Fu• Yihuan Xu• Wei Wang (NJIT) • Nicolle Clements• Ibrahim Turkoz• Elizabeth Stone• Bhramori Banerjee• Li He• Tingting Zhan• Vishwanath Iyer

SERVICE CONTRIBUTION

Department Service

- NTT reappointment and promotion committee. 2025
- Director of Graduate Programs. 2019-2023
- Departmental Great Teacher Award Committee. 2020
- Faculty Search Committee. 2019-2020
- Ph.D. Qualifying Exam Committee. 2011-present
- Committee of Joseph Heyse Publications Awards. 2011, 2012
- Department seminar coordinator. 2009-2010, 2010-2011, 2016-2017

School Service

- Evaluator of the Annual PhD Student Research Competition. 2025
- Merit Committee. 2013, 2024
- DPC committee. 2019-2023
- P&T Promotion Committee of RIHM. 2021
- MPC committee. 2019-2020
- Young Research Forum Evaluation Committee. 2020
- AACSB Faculty Committee. 2014

Professional Service

- Co-Editor-in-Chief, *Communications in Statistics, Theory and Methods.* 2025-present
- Associate editor, *The American Statistician.* 2024-present
- Associate editor, *Journal of Biopharmaceutical Statistics.* 2024-present
- Associate editor, *Statistical Analysis and Data Mining.* 2013-present
- Editorial board reviewers, *Journal of Machine Learning Research.* 2020-present
- Chair of the membership committee, International Chinese Statistical Association (ICSA). 2022-2024
- Chair of the Conference on Advances in Multiple Testing. 2023
- NSF review panel. 2020
- Program committee of *ICSA* Applied Statistics Symposium. 2018
- Program committee of *ICSA*-China Conference. 2019
- Membership committee, *ICSA*. 2015-2017
- Organizing committee of the conference on high dimensional statistics. 2013
- Ad hoc review (* more than once) for:

  • Scandinavian Journal of Statistics • Journal of the American Statistical Association(*) • Journal of the Royal Statistical Society, Series B(*) • Biometrika • Annals of Statistics • Biometrics • The American Statistician(*) • Statistica Sinica(*) • Journal of Statistical Planning and Inference(*) • Journal of the Korean Statistical Society • Statistical Analysis and Data Mining • Journal of Probability and Statistics • Journal of Nonparametric Statistics • Journal of Statistical Modeling • Test • Journal of Biopharmaceutical Research • Statistics and Probability Letters.

PRESENTATIONS

50+ conference and department talks.

Presentation at Conferences and Meetings

- Optimization-Based Framework for Nonparametric Empirical Bayes Inference. 6th US-Japan Workshop on Bayesian Statistics 2025
- Optimization-Based Framework for Nonparametric Empirical Bayes Inference. Statistics + AI Frontier Summit. 2025
- Optimization-Based Framework for Nonparametric Empirical Bayes Inference. 8th International Conference on Econometrics and Statistics 2025
- Testing of Independence and Exchangeability using Binary Expansion Testing. Keystone State Statistics Symposium. 2024

- On the testing of multiple hypotheses in sliced inverse regression. 7th International Conference on Econometrics and Statistics 2024
- On the testing of multiple hypotheses in sliced inverse regression. ICSA-China 2024
- Bayesian Analysis of Multiple Index Additive Models. Joint Statistical Meetings. 2023
- On the testing of multiple hypotheses in sliced inverse regression. ICSA Symposium. 2023
- Model-free Multiple Testing using Mirror Statistics (MMM). CMStatistics. 2022
- BEAUTY Powered BEAST. 12th International Conference on Multiple Comparison Procedures. 2022
- BEAUTY Powered BEAST. ICSA-China Conference. 2022
- BEAUTY Powered BEAST. ICSA Symposium. 2022
- BEAUTY Powered BEAST. 63rd ISI World Statistics Congress 2021 (ISI WSC 2021). 2021
- Bayesian selective inference, International Seminar on Selective Inference. 2020
- Controlling False Discovery Rate Using Gaussian Mirrors. CMStatistics. 2020
- Global testing under the sparse alternatives for single index models, CMStatistics Conference, Pisa, Italy, 2018
- Global testing under the sparse alternatives for single index models, The Second Annual Meeting of International Consortium of Chinese Mathematicians, Taipei, Taiwan, 2018
- Group Assisted Multiple Testing. ICSA Applied Statistics Symposium, New Jersey 2018
- Group Assisted Multiple Testing. Conference on Statistical Learning and Data Science. Chapel Hill, NC 2016
- Rate optimal multiple testing procedure in high-dimensional regression, IMS China International Conference on Statistics and Probability, Yunnan, China 2015
- Rate optimal multiple testing procedure in high-dimensional regression, ICSA-China Statistics Conference, Shanghai, China 2015
- Rate optimal multiple testing procedure in high-dimensional regression, Joint Statistical Meetings, Boston, MA 2014
- Rate optimal multiple testing procedure in high-dimensional regression, International Workshop on Multiplicity, Shanghai, China 2014
- Optimal Multiple Testing Procedure Under Linear Regression Model, ENAR Conference, Orlando, FL 2013
- Optimal Multiple Testing Procedure Under Linear Regression Model, Joint Statistical Meetings, San Diego, CA 2012
- Capturing the Severity of Type II Errors in High-Dimensional Multiple Testing, ICSA Applied Statistics Symposium, Boston, MA, 2012

- On the Credible Interval under the Zero-Inflated Mixture Prior in High Dimension Inference, Joint Statistical Meetings, Miami, FL, 2011
- On the Generalized Benjamini-Hochberg procedure, IMS-China International Conference on Statistics and Probability, Xi'an, Shanxi, 2011
- On the Generalized Benjamini-Hochberg procedure, ICSA Applied Statistics Symposium, New York City, NY, 2011
- Empirical Bayes Confidence Intervals for Selected Parameters for a Large Number of Normal Populations with Unequal but Estimable Means and Variances, the Eighth ICSA International Conference, Guangzhou, Guangdong, 2010
- Empirical Bayes Confidence Intervals for Selected Parameters for a Large Number of Normal Populations with Unequal but Estimable Means and Variances, Joint Statistical Meetings, Vancouver, Canada, 2010
- Empirical Bayes FCR Controlling Confidence Interval, Joint Statistical Meetings, Washington DC, 2009
- Empirical Bayes Confidence Intervals Shrinking Both Means and Variances, the 17th annual International Chinese Statistical Association Applied Statistics Symposium, Piscataway, NJ, 2008

Departmental Colloquia

- Testing of Independence and Exchangeability using Binary Expansion Testing. Virginia Polytechnic Institute and State University. 2024
- False Discovery Rate Control via Mirror Statistics. University of Georgia. 2023
- False Discovery Rate Control via Mirror Statistics. Pennsylvania State University. 2023
- On the testing of multiple hypotheses in sliced inverse regression. University of Pittsburgh. 2022
- Model-free Multiple Testing using Mirror Statistics (MMM). Colorado State University. 2022
- BEAUTY powered BEAST. University of California at Riverside. 2022
- Controlling False Discovery Rate Using Gaussian Mirrors. North Carolina State University. 2020
- Global testing under the sparse alternatives for single index models. George Mason University. 2019
- Nonparametric Empirical Bayes Estimator For Simultaneous Variances, University of Connecticut, CT. 2017
- A New Approach to Multiple Testing of Grouped Hypotheses, University of Illinois Urbana-Champaign, IL. 2016
- A New Approach to Multiple Testing of Grouped Hypotheses, Purdue University, IN. 2016

- A New Approach to Multiple Testing of Grouped Hypotheses, University of Delaware, DE. 2015
- Testing Multiple Hypotheses in Big Data Analysis, Southwest Jiaotong University, Chengdou, China. 2015
- Testing Multiple Hypotheses in Big Data Analysis, Beihang University, Beijing, China. 2015
- Optimal Multiple Testing Methods, Auburn University, AL. 2014
- Optimal Multiple Testing Procedure Under Linear Regression Model, Department of Statistics, University of Georgia, GA. 2013
- Empirical Bayes Confidence Intervals for Selected Parameters in High dimension, National Institute of Environmental Health Sciences (NIEHS), Research Triangle Park, NC. 2011
- On the Credible Interval under the zero-inflated Mixture Prior in High Dimension Inference, Department of Mathematics, New Jersey Institute of Technology, Newark, NJ. 2011
- Empirical Bayes Confidence Intervals for Selected Parameters for a Large Number of Normal Populations with Unequal but Estimable Means and Variances, Department of Operations Research and Financial Engineering, Princeton University, Princeton, NJ. 2011
- Empirical Bayes Confidence Intervals for Selected Parameters with Unknown and Unequal Variances, Department of Statistics, University of Missouri, Columbia, MO. 2010
- Empirical Bayes Confidence Intervals for Selected Parameters with Unknown and Unequal Variances, Department of Statistics, Nankai University, Tianjin, China. 2010
- Empirical Bayes Confidence Intervals Shrinking Both Means and Variances, Department of Statistics, Temple University, Philadelphia, PA. 2009
- Empirical Bayes Confidence Intervals Shrinking Both Means and Variances, Department of Mathematics, Syracuse University, Syracuse, NY. 2008

SOFTWARE

1. R package: Binary Expansion Test (BET)
2. R package: generalized signal-to-noise ratio (gSNR)
3. R package: CLAT: Cdf and Local fdr Assisted multiple Testing method (CLAT)
4. R package: LassoSIR: Sparse Sliced Inverse Regression via Lasso (LassoSIR)
5. R package: Multiple Testing Procedure for Grouped Hypotheses (GroupTest)
6. Bayesian LASSO with Zero Inflated Mixture Prior
7. EBVariant: An Empirical Bayes testing procedure for detecting variants in analysis of next generation sequencing data.

Last updated: November 2025.